

## 1 Design frozen before any outcome

### Protocol registered and hashed

null model · permutations · seed · weightings  
analysis aborts on a hash mismatch

### Thresholds set in advance

4 verdicts named before any result  
no threshold may be added or moved later

### Sensitivity panel declared

7 analyses fixed in the protocol  
including the IS family named ahead of time

## 2 Genome-wide census, verified independently

### ISEScan 1.7.3, pinned environment

HMMER 3.3.2 · BLAST+ 2.17.0 · FragGeneScan 1.32  
one chromosome at a time

### 190,999 structural elements

145,779 complete · 6,190 chromosomes  
one status per chromosome, none dropped

### 36 independent re-runs

seeded selection written before the first re-run  
36 of 36 reproduced the recorded count

## 3 Outcome sealed until the cohort was fixed

### COHORT LANE

#### Cohort built

admission rules applied to every occurrence  
no outcome column read

#### Cohort receipt on disk

counts · species · families · exclusions  
written before any estimate exists

#### Null model run

relocation within each chromosome only  
B = 2,000 · seed 20260824

  
**OUTCOME SEALED**

*outcome inaccessible while the cohort was being fixed*

### OUTCOME LANE

#### Observed distances held unread

no species outcome computed  
no threshold adjustable

#### Outcome read

only after the cohort receipt was written  
gates then applied mechanically

## 4 Gates and independent verification

### 4 gates applied mechanically

enrichment · comparators · balancing · clusters  
thresholds unchanged from registration

### Separate code path

every mechanic re-implemented brute force  
20 of 20 checks pass · 400/400 distances

### Verdict

one of 4 options registered in advance  
thresholds were never revisited

**BACKGROUND\_NORMALISED\_HOST\_CONDITIONING\_SUCCESS**